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ORIGINAL ARTICLE



## Significant impact of antibiotic exposure on GI-GVHD, NRM, and GRFS following allogeneic HCT with non-myeloablative Flu-TBI conditioning

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### ABSTRACT

**Background:** Acute gastro-intestinal graft-versus-host disease (GI-GVHD) and non-relapse mortality (NRM) after allogeneic HCT are closely related to loss of microbial diversity and intestinal dominance by single taxa resulting from the use of antibiotics, dietary changes, and mucosal barrier injury. There is a paucity of data on the impact of use of antibiotics in HCT after Flu-TBI-based non-myeloablative (NMA) conditioning where there is absence of mucositis and limited malnutrition.

**Methods:** We did a retrospective single-center analysis of patients receiving Flu-TBI-based NMA HCT for a high-grade myeloid malignancy, mostly AML, and MDS, or acute lymphoblastic leukemia (ALL). We analyzed the impact of pre-engraftment antibiotic exposure, prophylactic ciprofloxacin, and or treatment with broad-spectrum cephalosporin/carbapenem, on HCT outcomes, with a focus on the incidence of acute GI-GVHD by day 180 and NRM at 1 year.

**Results:** A total of 150 patients were evaluable with a median age of 62 years. Antibiotics were used in 90 patients; 60 prophylactic use only and 30 therapeutic use with or without previous prophylaxis. Antibiotic use resulted in a significant higher incidence of GI-GVHD Stage 1–4; 29% (26/90) versus 5% (3/60) in those not receiving antibiotics (OR 8.1 (95% CI 2.3–28.3),  $p=0.001$ ). Use of antibiotics resulted in higher 1-year NRM (19% (17/90) versus 10% (6/60), HR 2.3,  $p=0.06$ ), and decreased 2-year GRFS (42% (38/90) versus 55% (33/60), HR 1.7,  $p=0.04$ ), but did not impact RFS or OS.

**Conclusions:** Use of antibiotics was related to the occurrence of GI-GVHD, NRM, and GRFS in patients receiving truly NMA HCT. Therefore, in the absence of mucositis and low incidence of bacteremia, antibiotics can and should be used restrictively in this setting.

### ARTICLE HISTORY

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Antibiotics; microbial diversity; allogeneic stem cell transplantation; gastro-intestinal graft-versus-host disease; non-myeloablative conditioning

### Introduction

Acute gastro-intestinal graft-versus-host disease (GI-GVHD) remains one of the major causes of mortality after allogeneic hematopoietic cell transplantation (HCT) [1]. Recent work has identified changes in the gut microbiota, especially the loss of microbial diversity and domination of single taxa, as a key modulator of acute GI-GVHD and the associated non-relapse mortality (NRM) [2,3]. The exact mechanisms underlying alterations in the microbiota leading to acute GI-GVHD are still incompletely understood, but several factors have been identified. These include amongst others decreased food intake, mucosal barrier injury with decreased barrier function, transit time and mucus composition, and

use of drugs especially antibiotics [4]. Antibiotics, have a profound impact on the intestinal microbiota and their metabolites, e.g. butyrate, which normally promote intestinal tissue homeostasis and immune tolerance post-allogeneic HCT [5]. While in the early phase after allogeneic HCT antibiotics can reduce the incidence of mainly gram-negative bloodstream infections [6], early antibiotic treatment results in higher NRM and worse OS compared to patients with late or no additional broad-spectrum antibiotics [3,7–11].

Current data are mostly from the setting of reduced-intensity (RIC) and myeloablative (MAC) HCT. There is a lack of data from truly non-myeloablative (NMA) HCT [12] and in the largest study thus far by Peled et al. [3] NMA HCT was clearly underrepresented

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(<10%). MAC and RIC HCT are characterized by the occurrence of (severe) mucosal barrier injury, i.e. oral and intestinal mucositis, which is associated with a decreased mucosal barrier function and mucus composition by which they disrupt the intestinal microbiota [12]. The induction of mucositis is therefore a potential confounder in the effort of determining the impact of antibiotics. Truly NMA HCT regimens are considered non-mucotoxic and result in less mucositis, bloodstream infections, and malnutrition. Therefore, we performed a retrospective analysis in this particular setting trying to determine the sole impact of antibiotic use on GI-GVHD and HCT outcomes.

## Patients and methods

### Patients

We performed a single-center retrospective analysis in 176 consecutive patients receiving allogeneic HCT following Flu-TBI-based NMA conditioning for a hematological malignancy at the Radboud University Medical Center Nijmegen between February 2013 and December 2020. Only patients with a follow-up of at least 6 months and HCT for a high-grade myeloid malignancy, mostly acute myeloid leukemia (AML) and myelodysplastic syndrome with excess of blasts (MDS-EB), or acute lymphoblastic leukemia (ALL), were included leaving 150 evaluable patients.

At the time of treatment, all patients provided informed consent for the prospective collection of data and samples for investigational use. This retrospective study was approved by the Institutional Review Board at Radboud University Medical Center Nijmegen.

### Conditioning regimen and post-HCT immunosuppression

Patients received Flu-TBI conditioning consisting of 30 mg/m<sup>2</sup> fludarabine on days -4, -3, and -2, and low-dose TBI (2 Gy) on day -1. In case of a mismatched unrelated donor (MMUD) additional *in vivo* T cell depletion with 2 mg/kg/day Thymoglobulin (rabbit-ATG; Genzyme) was given on days -8, -7, -6, and -5 [13]. In accordance with study protocols (NCT02252107), some patients with adverse risk myeloid malignancies also received 10 d of decitabine (20 mg/m<sup>2</sup>) [14]. On day 0, all patients received a T cell-replete graft with mobilized peripheral blood stem cells or bone marrow stem cells.

Post-HCT immunosuppression consisted of mycophenolate mofetil for 28 or 96 d for matched related

donors (MRDs) and unrelated donors (MUD and MMUD), respectively, and oral cyclosporine A (CPS) for 180 d. CPS started at day -3 and was initially dosed at 4.5 mg/kg twice daily, with tapering starting on day +100.

### Treatment and prophylaxis for bacterial infections

In our center, the use of antibiotics in Flu-TBI-based HCT was not protocolized and left at the discretion of the treating physician. This with the exception of those patients receiving decitabine in their conditioning, where ciprofloxacin prophylaxis was standard. Ciprofloxacin 500 mg BID was given orally from start of the conditioning regimen until occurrence of febrile neutropenia or neutrophil recovery. In case of febrile neutropenia, determined as a temperature  $\geq 38.5^{\circ}\text{C}$  and neutrophils  $< 0.5 \times 10^9/\text{L}$ , ceftazidime 2000 mg TID was started as empiric first-line therapy in most of the cases. Patients with known colonization or previous infections with bacteria resistant to ceftazidime were initially treated with meropenem. Patients who had a positive serology test for herpes simplex virus and/or varicella zoster virus received 500 mg valacyclovir BID. Antifungal prophylaxis was not routinely given during the conditioning treatment or neutropenic phase. After engraftment, patients received *Pneumocystis jirovecii* prophylaxis consisting of cotrimoxazole 480 mg QD.

### Definitions

Acute GVHD was defined and graded according to the criteria proposed by Harris et al. [15]. Clinically, suspected cases of acute GVHD were confirmed histologically. The incidence of acute GVHD was determined at day +180 because with the use of Flu-TBI-based conditioning acute GVHD occurs beyond day 100 in a considerable proportion of patients [13,16]. The occurrence of NRM, relapse-free survival (RFS), and overall survival (OS) were defined in accordance with EBMT statistical guidelines. GVHD and relapse-free survival (GRFS) was defined as survival without relapse, acute GVHD grade III-IV or severe chronic GVHD [17]. The disease risk index (DRI) and HCT-CI were determined in accordance with published scoring systems [18,19].

Patients were classified, according to their pre-engraftment exposure to antibiotic, into two subgroups; patients not receiving antibiotics (AB no) and patients receiving any antibiotics (AB all). The latter group consisted of patients receiving antibacterial prophylaxis only (AB(p)) or therapeutic broad-spectrum

antibiotics (AB(t)), i.e. empiric treatment of febrile neutropenia or directed therapy for a microbiologically or clinically defined infection, with or without prior prophylaxis. Anaerobic broad-spectrum antibiotics consisted of meropenem or ceftazidime/metronidazole combination and non-anaerobic broad-spectrum antibiotics consisted of ceftazidime alone.

### Statistical analysis

We used descriptive statistics to analyze the patient, donor, and HCT characteristics. Differences between categorical and continuous variables by Chi-Square and independent *t*-test, respectively. The Kaplan–Meier method was used to estimate OS, RFS, and GRFS. To examine the impact of competing risks, we performed a sensitivity analysis for the incidence of GVHD outcome excluding patients dying before day 180. Putative factors affecting severe acute GVHD, GI-GVHD, NRM, OS, RFS, and GRFS were selected for univariate analysis. Considering the small cohort with a limited number of events only factors with a *p* value < 0.2 were incorporated in a multivariate analysis using logistic regression or Cox regression. Because of small numbers in the statistical analysis the subgroups ‘anaerobic and non-anaerobic broad-spectrum antibiotics’ were merged. Analyses were done with IBM SPSS Statistics version 25 (IBM Corp., Armonk, NY). Differences with a *p* value of < 0.05 were considered statistically significant.

### Results

In the final analysis, 150 patients were included that had received Flu-TBI-based NMA conditioning. The characteristics and features of the patient, donors, and HCT procedures are summarized in Table 1. Median age was 59 years and 72 (53%) were male. Most patients received a HCT for AML and MDS-EB and were in CR1. Roughly one-third of patients received decitabine in addition to Flu-TBI and ATG was given for MMUD in 20 patients.

#### Antibiotic exposure and infectious complications

Antibiotic exposure occurred in 90 patients (60%) with the majority receiving prophylactic ciprofloxacin only, 60 (40%), and with 30 (20%) requiring broad spectrum antibiotics. The indication for broad spectrum antibiotic use was empirical therapy for fever during neutropenia in all 30 patients. In five patients antibiotic therapy with gram-positive coverage was added based

on the occurrence of BSI (*N*=4) or clinically defined infection (*N*=1), i.e. neutropenic enterocolitis. The incidence of BSI was very low (4/150; 2.5%); cultured micro-organisms consisted of *Corynebacterium jeikeium*, *Bacillus cereus*, *Staphylococcus hominis*, and *Enterococcus faecium*.

The use of antibiotics was as expected not equally distributed amongst conditioning types. Patients receiving decitabine in their conditioning experienced neutropenia more often and with longer duration. Almost all of these patients received prophylaxis (96.5%; 55/57) compared to those without decitabine (37.6%; 35/93), *p*=0.001. Empirical BS AB was also applied more often in those receiving decitabine (26.3% (15/57) versus 16.1% (15/93)), *p*=0.1.

#### Graft-versus-host disease

In our patient cohort, the median time of acute GVHD grade II–IV was 78 d (IQR 33–132) and of GI-GVHD Stage 1–4 was 84 d (33–128). The incidence of acute GVHD at day 180 was 34% (51/150) and did not differ with the use of antibiotics or not, but with CPS through levels and donor type (Table 2, Supplementary Table 1). Severe acute GVHD occurred in 15% (23/150), and occurred significantly more often in those using antibiotics; 21% (19/90) versus 7% (4/60), HR 4.1 (95% CI 1.1–15.0), *p*=0.03 (Table 2). Occurrence of severe GVHD was mainly due to GI-GVHD and in total thirty patients (20%) developed GI-GVHD. As expected GI-GVHD was significantly more common with use of antibiotics; 30% (27/90) versus 5% (3/60), HR 9.6 (95% CI 2.5–37.5), *p*=0.001 (Figure 1, Table 2).

A sensitivity analysis on the incidence of acute and chronic GI-GVHD leaving out patients experiencing death (before day 180) did not show any difference (Supplementary Table 1(A,B)). The impact of antibiotic use seemed to be gradual with an increasing negative impact with the use of prophylactic ciprofloxacin, non-anaerobic broad-spectrum therapeutic antibiotics, and anaerobic broad-spectrum therapeutic antibiotics (Figure 1). Use of ATG, decitabine, and CsA trough levels had no significant impact on multivariate analysis (Supplementary Table 2).

As most patients receiving decitabine were exposed to ciprofloxacin, conditioning itself could be an important confounder. However, in multivariate analysis the impact of antibiotic exposure remained significant (Supplementary Table 2). Nevertheless, we repeated our analysis in those only receiving Flu-TBI and found

**Table 1.** Characteristics of patients, donors, and HCT procedure characteristics.

|                                                      | Antibiotic use (n=90) | No antibiotics (n=60) | p Value |
|------------------------------------------------------|-----------------------|-----------------------|---------|
| Male sex, no (%)                                     | 47 (52%)              | 33 (55%)              | Ns      |
| Age, no (%)                                          |                       |                       |         |
| • <60 years                                          | 37 (41%)              | 17 (28%)              | 0.1     |
| • ≥ 60 years                                         | 53 (59%)              | 43 (72%)              |         |
| Diagnosis, no (%)                                    |                       |                       | Ns      |
| • AML                                                | 56 (62%)              | 42 (70%)              |         |
| • MDS-EB (1 or 2)                                    | 26 (29%)              | 10 (17%)              |         |
| • ALL/LBL                                            | 3 (3.5%)              | 3 (5%)                |         |
| • Other*                                             | 5 (5.5%)              | 5 (8%)                |         |
| Disease status at HCT, no (%)                        |                       |                       | Ns      |
| • CR1                                                | 84 (93%)              | 55 (92%)              |         |
| • Other: CR2, PR, upfront                            | 6 (7%)                | 5 (8%)                |         |
| Disease risk index (DRI), no (%)                     |                       |                       | 0.2     |
| • Low                                                | 2 (2%)                | 2 (3%)                |         |
| • Intermediate                                       | 57 (63%)              | 42 (70%)              |         |
| • High/very-high                                     | 31 (35%)              | 16 (27%)              |         |
| HCT-CI, no (%)                                       |                       |                       | Ns      |
| • 0–2                                                | 37 (41%)              | 28 (47%)              |         |
| • 3+                                                 | 53 (59%)              | 32 (53%)              |         |
| Donor, no (%)                                        |                       |                       | Ns      |
| • MRD                                                | 23 (25%)              | 19 (32%)              |         |
| • MUD                                                | 53 (59%)              | 35 (58%)              |         |
| • MMUD                                               | 14 (16%)              | 6 (10%)               |         |
| Graft source, no (%)                                 |                       |                       | Ns      |
| • PBSC                                               | 88 (98%)              | 59 (98%)              |         |
| Conditioning regimen, no (%)                         |                       |                       | 0.001   |
| • Flu-TBI                                            | 35 (39%)              | 58 (97%)              |         |
| • Dec-Flu-TBI                                        | 55 (61%)              | 2 (3%)                |         |
| Use of ATG, no (%)                                   | 14 (16%)              | 6 (10%)               | Ns      |
| CMV status recipient, no (%)                         |                       |                       | Ns      |
| • Negative                                           | 26 (29%)              | 19 (32%)              |         |
| • Positive                                           | 64 (71%)              | 41 (68%)              |         |
| Neutrophils, no (%)                                  |                       |                       | 0.001   |
| • < 0.25 × 10 <sup>9</sup> /L                        | 76 (84%)              | 32 (53%)              |         |
| • ≥ 0.25 × 10 <sup>9</sup> /L                        | 14 (16%)              | 28 (47%)              |         |
| Median neutrophil nadir (IQR)                        | 0.02 (0.01–0.1)       | 0.24 (0.01–0.5)       |         |
| Cyclosporine A trough concentration week 1–4, no (%) |                       |                       | 0.03    |
| • < 300 ng/mL                                        | 24 (27%)              | 7 (12%)               |         |
| • ≥ 300 ng/mL                                        | 66 (73%)              | 53 (88%)              |         |
| Antibiotics, no (%)                                  |                       |                       |         |
| • Prophylactic                                       | 60 (67%)              | NA                    |         |
| • Treatment                                          | 30 (33%)              | NA                    |         |

Differences between categorical and continuous variables by Chi-square and independent t-test, respectively.

AML: acute myeloid leukemia; MDS-EB: myelodysplastic syndrome with excess blasts; ALL: acute lymphoblastic leukemia; LBL: lymphoblastic lymphoma; PBSC: peripheral blood stem cells; MRD: matched related donor; MUD: matched unrelated donor; MMUD: mismatched unrelated donor; Flu-TBI: fludarabine-total body irradiation; Dec-Flu-TBI: decitabine-fludarabine-total body irradiation; ATG: anti-thymocyte globulin; GVHD: graft-versus-host disease; GI-GVHD: gastro-intestinal graft-versus-host disease

\*MPN-BP, CML-BP, atypical CML, and CMML 1 or 2.

the same impact; incidence GI-GVHD 34% (12/35) in those with *versus* 5% (3/58) in those without antibiotics,  $p=0.001$  (Supplementary Figure S1).

Chronic GVHD with NIH moderate and severe intensity occurred in 44 patients (29%) and did not differ amongst groups classified by antibiotic exposure. There was some difference in severe chronic GVHD, 17% (15/90) in those with *versus* 10% (6/60) in those without antibiotics, but this was not statistically significant (Table 2).

### HCT outcome

Median follow-up post HCT of surviving patients was 32 months (range 5–62). The OS, RFS, and GRFS at

2 years were 65%, 60%, and 47%, respectively (Supplementary Figure S2). The NRM at 1 year was 15.3% and differed between those receiving antibiotics or not; 19% (17/90) *versus* 10% (6/60), HR 2.3 (95% CI 0.95–5.8), although in multivariate analysis this was only borderline significant;  $p=0.06$  (Figure 2(A), Supplementary Table 2). Both OS and RFS were not significantly different between those exposed to antibiotics or not and mostly DRI impacted these outcomes (Supplementary Table 2, Figure S3). GRFS was significantly impacted by DRI, however also by the use of antibiotics; 42% (38/90) *versus* 55% (33/60), HR 1.7 (95% CI 1.1–3.0),  $p=0.04$ , Supplementary Table 2 and Figure 2(B). This difference was mainly the result of the higher incidence of severe acute GVHD, with GI



involvement. The use of decitabine, ATG, and CsA trough levels did not impact GRFS on multivariate analysis.

Causes of death were very different for patients receiving antibiotics with a high incidence of acute GVHD-related mortality (35%) whereas mortality was predominantly relapse-related in patients without use of antibiotics (72%) (Table 3).

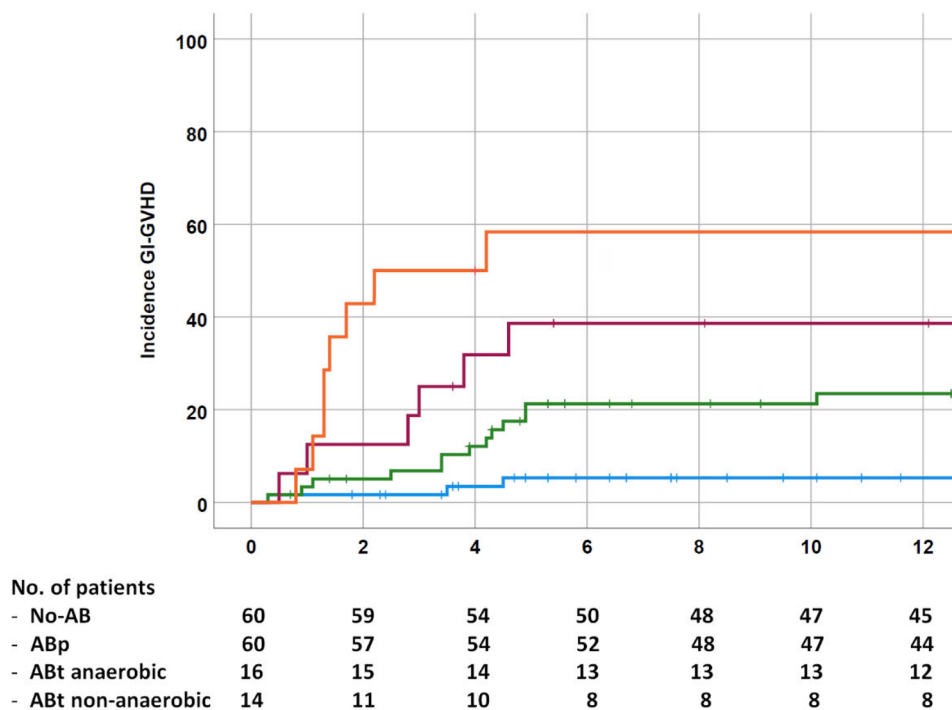
**Table 2.** Incidence of acute and chronic GVHD by antibiotic exposure.

|               | Antibiotic use<br>(n=90) (%) | No antibiotics<br>(n=60) | p Value |
|---------------|------------------------------|--------------------------|---------|
| Acute GVHD    |                              |                          | ns      |
| Grade II–IV   |                              |                          |         |
| • No          | 57 (63)                      | 42 (70)                  |         |
| • Yes         | 33 (37)                      | 18 (30)                  |         |
| Acute GVHD    |                              |                          | 0.02    |
| Grade III–IV  |                              |                          |         |
| • No          | 71 (79)                      | 56 (93)                  |         |
| • Yes         | 19 (21)                      | 4 (7)                    |         |
| GI-GVHD Stage |                              |                          | 0.001   |
| I–IV, no (%)  |                              |                          |         |
| • No          | 63 (70)                      | 57 (95)                  |         |
| • Yes         | 27 (30)                      | 3 (5)                    |         |
| Chronic GVHD, |                              |                          | ns      |
| no (%)        |                              |                          |         |
| • Moderate    | 12 (13)                      | 11 (18)                  |         |
| • Severe      | 15 (17)                      | 6 (10)                   |         |

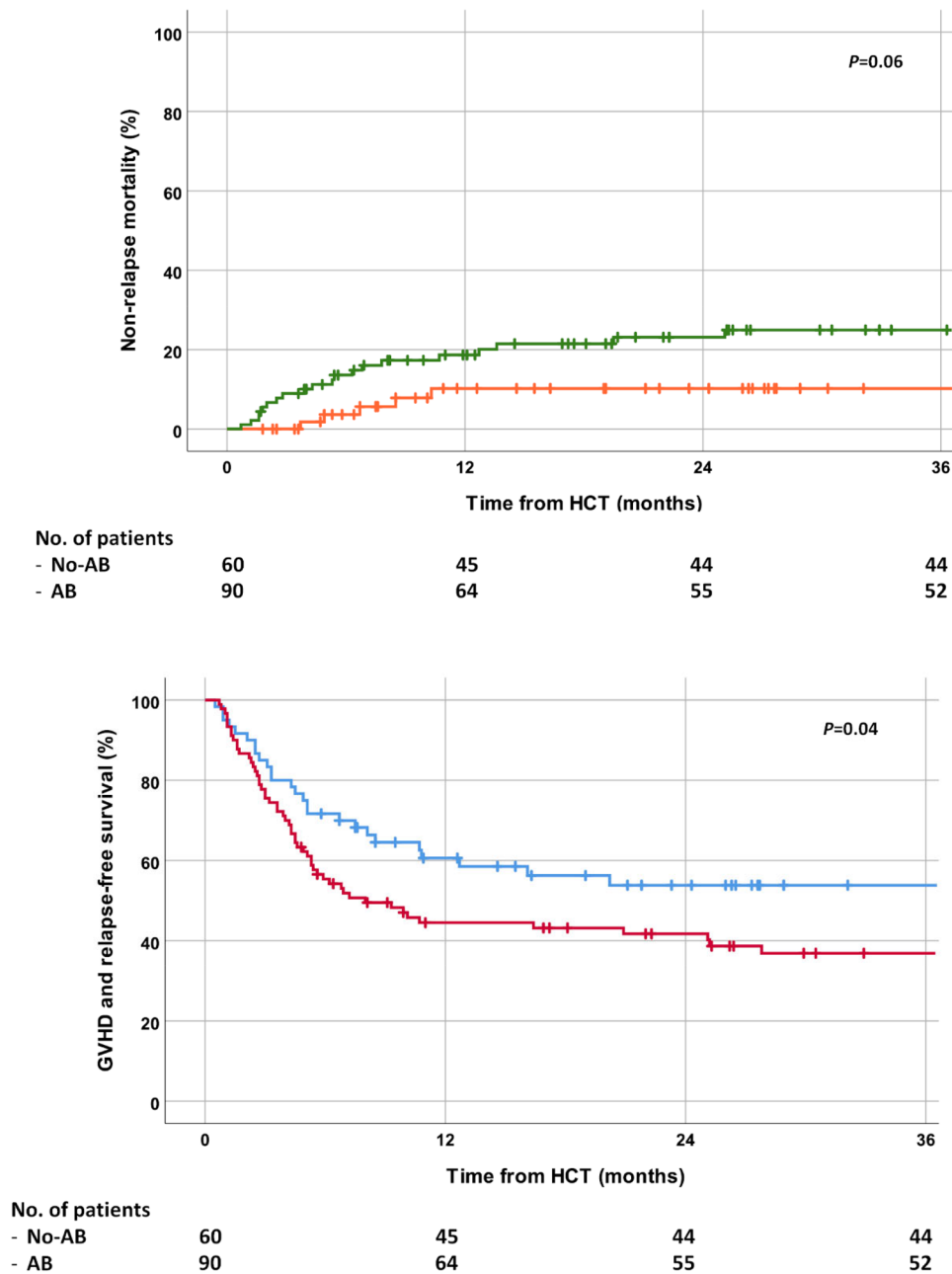
\*A sensitivity analysis excluding patients dying before day +180 confirmed the robustness of these values.

## Discussion

In this retrospective study in truly NMA HCT recipients we demonstrate that the use of antibiotics (prophylactic small spectrum and or therapeutic broad spectrum) is associated with an increased incidence of acute GI-GVHD, NRM, and GRFS. However, we did not demonstrate an impact on RFS and OS. As shown in Figure 1, there was a gradual increased risk for developing GI-GVHD with the use of prophylactic ciprofloxacin to more broad-spectrum antibiotics, and especially those with anaerobic activity, such as meropenem or ceftazidime/metronidazol. This suggests that there is an incremental risk with the use of more and more broad spectrum antibiotics. Earlier studies have shown that not all antibiotics are equal with regards to the negative effect on the gut microbiota and consequently their impact on GI-GVHD and NRM, with the largest impact of antibiotics that disrupt commensal anaerobic taxa. Therefore, ciprofloxacin and ceftazidime have been designated rather safe [20], although controversy exists on the question if really commensal sparing antibiotics exist [6–9]. Nevertheless, our clinical data suggest that in the context of Flu-TBI conditioning with T-cell replete HCT exposure to ciprofloxacin and ceftazidime clearly had an impact on HCT outcomes, predominantly GI-GVHD. Regretfully correlative fecal



**Figure 1.** The incidence of gastro-intestinal acute GVHD by antibiotic exposure. Incidence of Stage 1–4 GI-GVHD in those with no antibiotic exposure (blue, N=60) versus those with prophylactic antibiotics (green, N=60), non-anaerobic broad-spectrum antibiotics (red, N=16), and anaerobic broad-spectrum antibiotics (orange, N=14).



**Figure 2.** Outcomes of Flu-TBI based NMA HCT by antibiotic exposure. A) Difference in non-relapse mortality in those with (green) and without antibiotic use (orange); borderline statistically significant on multivariate analysis. B) GVHD-relapse free survival in those with (red) and without antibiotic use (blue);  $p=0.04$ .

samples were not collected in our study, so associations between antibiotic exposure and microbiome diversity could not be studied. However, our findings correspond with recent data showing a significant impact of use of fluoroquinolones on the treatment outcomes of immune checkpoint inhibitors in solid cancers where changes in the gut microbiota species including *Faecalibacterium*, *Ruminococcus*, and *Bifidobacteria*, are implicated [21,22].

We believe that the context of NMA conditioning is of particular interest and importance because in

microbiota analysis in RIC and MAC HCT, mucositis and malnutrition, are potential confounders. Mucositis is a major cause of fever during neutropenia and strongly related to use of antibiotics making these two 'inseparable entities'. In addition, malnutrition comes with more severe oral and intestinal mucositis and their particular impact on the microbiota cannot be determined. Flu-TBI is a truly NMA conditioning with virtually absent oral and intestinal mucositis, i.e. citrulline levels remain above the  $10\mu\text{mol/L}$  threshold that defines gut failure (Supplementary Figure S4). In addition, malnutrition in

**Table 3.** Causes of death.

|                                 | Antibiotic use<br>(n = 90) | No antibiotics<br>(n = 60) |
|---------------------------------|----------------------------|----------------------------|
| Died                            | 34 (38%)                   | 18 (30%)                   |
| Cause of death                  |                            |                            |
| • Relapse related               | 15 (44%)                   | 13 (72%)                   |
| • GVHD related                  | 12 (35%)                   | 2 (11%)                    |
| • Rejection/graft failure       | 2 (6%)                     | 1 (6%)                     |
| • Infection                     | 2 (6%)                     | 1 (6%)                     |
| • Idiopathic pneumonia syndrome | 1 (3%)                     | 1 (6%)                     |
| • Secondary malignancy          | 2 (6%)                     | 0                          |

this setting is rare and mostly related to GI adverse effects of calcineurin inhibitors and MMF rather than malabsorption. So, this underlines that use of antibiotics was probably the main cause of the increased risk for GI-GVHD that we found in our analysis.

The use of decitabine was a potential confounding factor, because more profound neutropenia could be associated with a higher incidence of GI-GVHD as previously shown in Flu-TBI HCT [22]. Although decitabine is regarded non-mucotoxic at least theoretically effects of DNA hypomethylation might influence gut immunity. Hypomethylating agents increase the transcription of endogenous retroviruses (ERV) thereby eliciting interferon responses that recently have been linked to microbiota-induced inflammation, which might influence GI-GVHD pathogenesis [23,24]. Nevertheless, repeating the analysis with exclusion of the decitabine subgroup revealed the same significant impact of antibiotic use on development of GI-GVHD.

Use of antibiotics in HCT should be a decision weighing benefits and costs. In Flu-TBI HCT we believe that the costs are evident, but the benefits might be limited. Blood stream infections, especially with gram-negative bacteria, can be life-threatening complications in HCT, but are generally related to the occurrence of mucosal barrier injury. Therewith, use of fluoroquinolone prophylaxis can protect RIC and MAC HCT recipients from this complication as has been confirmed in multiple studies and meta-analysis showing reduced all-cause mortality [25]. Then again, in Flu-TBI, the occurrence of BSI is low, and mostly gram-positive bacteria are encountered. The fact that mucosal barrier integrity is maintained with this regimen limits the risk of gram-negative sepsis even in those developing neutropenia. Hence, we believe that there is no reason for standard fluoroquinolone prophylaxis in Flu-TBI conditioning, even when combined with decitabine, and even should be discouraged in order to preserve a healthy gut microbiota.

There are several limitations to our study. First of all, there are the natural drawbacks of retrospective single center studies including unmeasured confounders.

Second, antibiotic exposition during treatments preceding allogeneic HCT was not documented, so pre-existing dysbiosis could not be estimated. Finally, and most importantly, we had no data on gut microbiota composition and changes therein. This precludes a definite conclusion on causality and designation of specific taxa involved in the risk of GI-GVHD in the Flu-TBI setting.

In conclusion, we found a significant and incremental impact of antibiotic use on the incidence of GI-GVHD NRM and GRFS in recipients of truly NMA Flu-TBI-based HCT. NMA conditioning can be considered non-mucotoxic and antibiotic prophylaxis are therefore not required in this particular setting. Additional studies are necessary to define the microbiota composition and changes in Flu-TBI-based HCT to confirm our findings and depict the involved microbial taxa.

## Disclosure statement

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